Course code: MS 204
Course Name: Metals and Alloys
Pre-requisite: None
Objective of the course: To provide the fundamental understanding and applications of metals and alloys

Metals and Alloys

Unit-I (8 lectures)

Bonding in solids, structures of solids, structure of metals and alloys

Recapitulating the mechanical response of the metals and alloys, Mechanical properties: Stress-strain behavior, elastic and plastic deformation, Ferrous and Non-Ferrous alloys

Unit-II (12 lectures)

Advanced intermetallic alloys (fundamental and applications):
Martensite transformation: Types of solid-state phase transformation in metals and alloys, characteristics of martensite transformation taking Fe-C as example, concept of habit plane, lattice deformation shear and lattice invariant shear.

Martensitic transformation in Beta phase alloys, calculation of structure factor, stacking fault, peak shift.

Unit-III (8 lectures)
Geometrical compatibility, shape memory and magnetic shape memory, thermoelectric alloys. Alloys for solid state cooling (barocaloric, electrocaloric, magnetocaloric) applications,

Unit-V (12 lectures)
Advanced magnetic alloys (fundamental and applications): Heusler alloys and applications, Alloys for permanent magnets, superconducting alloys, Alloys for dental and bone replacement medical application. Magnetic multilayers, spin valve etc.

READINGS

TEXTBOOK:

1. Materials Science & Engineering: An Introduction, W.D. Callister. Jr.
2. Mechanical Metallurgy, G.E. Dieter.
3. Materials Science and Engineering, V. Raghavan.
4. Martensite transformation: Z. Nishiyama.
5. Shape Memory Materials: K. Otsuka, C. M. Wayman
6. Magnetism and Magnetic Materials: J. M. D. Coey
7. The Magnetocaloric Effect and its Applications: A. M. Tishin

REFERENCE BOOKS:

1. ASM Handbook of Metals Vol.8, 9th Edition
2. ASM Handbook of Alloy Phase Diagram Vol.3.